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SIMATS Online Programming Lab

Kanigasri M
192565010RD

JAVA PROGRAMMING

Classes, Objects and Inheritance

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QUESTIONS

Q1

Program based on Classes and Objects

5 marks

**Q2**

Initialize Object Data Using Setter Methods

5 marks

**Q3**

Student Details Using Default and Parameterized Constructors

10 marks

**Q4**

Net Salary Calculation Using Single Inheritance

10 marks

**Q5**

Q1 Program based on Classes and Objects

5 marks

Write a Java program to create a class named Student with the following data members:

int rollNo

String name

double marks

Create a method named displayDetails() to display the student's details.

In the main() method:

Create a Student object.

Read the values from the user.

Initialize the data members directly using the object (do not use a constructor or setter methods).

Invoke the displayDetails() method using the object.

Input Format

The first line contains an integer representing the roll number.

The second line contains the student's name.

Employee Gross Salary
Calculation Using Multilevel
Inheritance
10 marks

5/5 answered

The third line contains a decimal number representing the student's marks.

Output Format

Print the details in the following format:

Roll No: <rollNo>

Name: <name>

Marks: <marks>

Your Code

```
1 import java.util.Scanner;
2 class Student{
3     private int rollNo;
4     private String name;
5     private double marks;
6     void displayDetails(){
7         System.out.println("Roll No: "+rollNo);
8         System.out.println("Name: "+name);
9         System.out.println("Marks: "+marks);
10    }
11    public static void main(String[] args){
12        Scanner sc = new Scanner(System.in);
13        Student s = new Student();
14        s.rollNo = sc.nextInt();
15        sc.nextLine();
16        s.name = sc.nextLine();
17        s.marks = sc.nextDouble();
18        s.displayDetails();
19    }
```

```
20 }
```

Input

```
012  
sam  
92.9
```

Output

```
Roll No: 12  
Name: sam  
Marks: 92.9
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



5.00 / 5 marks

 Submitted



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10 marks

Q4

Net Salary Calculation Using Single Inheritance

10 marks

Q5

Q2 Initialize Object Data Using Setter Methods

5 marks

Write a Java program to create a class named Student with the following private data members:

`int rollNo``String name``double marks`

Create the following setter methods to initialize the data members:

`setRollNo(int rollNo)``setName(String name)``setMarks(double marks)`

Also, create a method named `displayDetails()` to display the student's details.

In the `main()` method:

Create a Student object.

Read the values from the user.

Initialize the data members using the setter methods only.

Employee Gross Salary
Calculation Using Multilevel
Inheritance
10 marks

5/5 answered

Call the `displayDetails()` method to display the student's information.

Input Format

The first line contains an integer representing the roll number.

The second line contains the student's name.

The third line contains a decimal number representing the student's marks.

Output Format

Display the details in the following format:

Roll No: <rollNo>

Name: <name>

Marks: <marks>

Your Code

```
1 import java.util.Scanner;
2 class Student{
3     private int rollNo;
4     private String name;
5     private double marks;
6     public void setRollNo(int rollNo){
7         this.rollNo = rollNo;
8     }
9     public void setName(String name){
10        this.name = name;
11    }
12    public void setMarks(double marks){
13        this.marks = marks;
14    }
15    public void displayDetails(){
16        System.out.println("Roll No: "+rollNo);
17        System.out.println("Name: "+name);
```

```
18         System.out.println("Marks: "+marks);
19     }
20     public static void main(String[] args){
21         Scanner sc = new Scanner(System.in);
22         Student s = new Student();
23         s.setRollNo(sc.nextInt());
24         sc.nextLine();
25         s.setName(sc.nextLine());
26         s.setMarks(sc.nextDouble());
27         s.displayDetails();
28         sc.close();
29     }
30 }
```

Input

```
012
sam
89.0
```

Output

```
Roll No: 12
Name: sam
Marks: 89.0
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



5.00 / 5 marks

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Student Details Using Default and Parameterized Constructors

10 marks

**Q4**

Net Salary Calculation Using Single Inheritance

10 marks

**Q5**

Q3 Student Details Using Default and Parameterized Constructors

10 marks

Write a Java program to create a class named Student with the following data members:

int rollNo

String name

double marks

Implement the following constructors:

Default Constructor

Initializes the data members with the following values:

Roll Number: 101

Name: Rahul

Marks: 85.5

Parameterized Constructor

Accepts the roll number, name, and marks as parameters and initializes the data members.

Create a method named displayDetails() to display the student's details in the following format:

Employee Gross Salary
Calculation Using Multilevel
Inheritance
10 marks

5/5 answered

Roll No: <rollNo>

Name: <name>

Marks: <marks>

Task

In the main() method:

Create the first Student object using the default constructor and display its details.

Read the roll number, name, and marks from the user.

Create the second Student object using the parameterized constructor and display its details.

Input Format

The first line contains an integer representing the roll number.

The second line contains the student's name.

The third line contains a decimal number representing the student's marks.

Output Format

Display the details of both students.

The first student's details should be displayed using the default constructor, followed by the second student's details using the parameterized constructor.

Your Code

```
1 import java.util.Scanner;
2 class Student{
3     int rollNo;
4     String name;
5     double marks;
6     Student() {
7         rollNo = 101;
8         name = "Rahul";
```

```
9         marks = 85.5;
10     }
11     Student(int rollNo,String name,double marks){
12         this.rollNo = rollNo;
13         this.name = name;
14         this.marks = marks;
15     }
16     void displayDetails(){
17         System.out.println("Roll No: "+rollNo);
18         System.out.println("Name: "+name);
19         System.out.println("Marks: "+marks);
20     }
21     public static void main(String[] args){
22         Scanner sc = new Scanner(System.in);
23         Student s1 = new Student();
24         s1.displayDetails();
25         int rollNo = sc.nextInt();
26         sc.nextLine();
27         String name = sc.nextLine();
28         double marks = sc.nextDouble();
29         Student s2 = new Student(rollNo,name,marks);
30         s2.displayDetails();
31         sc.close();
32     }
33 }
```

Input

```
102
sam
89.7
```

Output

```
Roll No: 101  
Name: Rahul  
Marks: 85.5  
Roll No: 102  
Name: sam  
Marks: 89.7
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

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10 marks

**Q4**

Net Salary Calculation Using Single Inheritance

10 marks

**Q5**

Q4 Net Salary Calculation Using Single Inheritance

10 marks

Write a Java program to demonstrate Single Inheritance.

Create a superclass named Employee with the following data members:

`int empId``String empName``double basicSalary`

Create a method named `getEmployeeDetails(int empId, String empName, double basicSalary)` to initialize the employee details.

Create a subclass named Salary that inherits from Employee.

In the subclass, create a method named `calculateSalary()` that calculates the following:

House Rent Allowance (HRA) = 20% of Basic Salary

Dearness Allowance (DA) = 10% of Basic Salary

Net Salary = Basic Salary + HRA + DA

Create another method named `displayDetails()` to display the employee details and the calculated salary.

Employee Gross Salary
Calculation Using Multilevel
Inheritance
10 marks

5/5 answered

Input Format

The first line contains an integer representing the Employee ID.

The second line contains the Employee Name.

The third line contains the Basic Salary.

Output Format

Display the output in the following format:

Employee ID: <empId>

Employee Name: <empName>

Basic Salary: <basicSalary>

HRA: <hra>

DA: <da>

Net Salary: <netSalary>

Your Code

```
1 import java.util.Scanner;
2 class Employee{
3     int empId;
4     String empName;
5     double basicSalary;
6     void getEmployeeDetails(int empId,String empName,double basicSalary)
7     {
8         this.empId = empId;
9         this.empName = empName;
10        this.basicSalary = basicSalary;
11    }
12 public static void main(String[] args){
13     Scanner sc = new Scanner(System.in);
14     Salary s = new Salary();
15     int id = sc.nextInt();
16     sc.nextLine();
```

```
16     String name = sc.nextLine();
17     double salary = sc.nextDouble();
18     s.getEmployeeDetails(id,name,salary);
19     s.calculateSalary();
20     s.displayDetails();
21     sc.close();
22 }
23 }
24 class Salary extends Employee{
25     double hra,da,netSalary;
26     void calculateSalary(){
27         hra = basicSalary*0.20;
28         da = basicSalary*0.10;
29         netSalary = basicSalary + hra + da;
30     }
31     void displayDetails(){
32         System.out.println("Employee ID: "+empId);
33         System.out.println("Employee Name: "+empName);
34         System.out.println("Basic Salary: "+basicSalary);
35         System.out.println("HRA: "+hra);
36         System.out.println("DA: "+da);
37         System.out.println("Net Salary: "+netSalary);
38     }
39 }
```

Input

```
012
sam
80000
```

Output

```
Employee ID: 12  
Employee Name: sam  
Basic Salary: 80000.0  
HRA: 16000.0  
DA: 8000.0  
Net Salary: 104000.0
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

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10 marks

**Q4**

Net Salary Calculation Using Single Inheritance

10 marks

**Q5**

Q5 Employee Gross Salary Calculation Using Multilevel Inheritance

10 marks

Write a Java program to demonstrate Multilevel Inheritance.

Create three classes as described below:

Class 1: Employee (Superclass)

Data Members:

int empId

String empName

Method:

getEmployeeDetails(int empId, String empName) – Initializes the employee details.

Class 2: Salary (Derived Class)

This class inherits from Employee.

Data Member:

double basicSalary

Method:

Employee Gross Salary
Calculation Using Multilevel
Inheritance
10 marks

5/5 answered

getBasicSalary(double basicSalary) – Initializes the basic salary.

Class 3: GrossSalary (Derived Class)

This class inherits from Salary.

Method:

calculateSalary() – Calculates:

House Rent Allowance (HRA) = 20% of Basic Salary

Dearness Allowance (DA) = 10% of Basic Salary

Gross Salary = Basic Salary + HRA + DA

Method:

displayDetails() – Displays the employee details and the calculated salary.

Input Format

The first line contains an integer representing the Employee ID.

The second line contains the Employee Name.

The third line contains the Basic Salary.

Output Format

Display the output in the following format:

Employee ID: <empId>

Employee Name: <empName>

Basic Salary: <basicSalary>

HRA: <hra>

DA: <da>

Gross Salary: <grossSalary>

Your Code

```
1 import java.util.Scanner;
```

```
2 class Employee{
3     int empId;
4     String empName;
5     void getEmployeeDetails(int id,String name){
6         empId=id;
7         empName=name;
8     }
9     public static void main(String[] args){
10         Scanner sc = new Scanner(System.in);
11         GrossSalary g = new GrossSalary();
12         g.getEmployeeDetails(sc.nextInt(),sc.next());
13         g.getBasicSalary(sc.nextDouble());
14         g.displayDetails();
15     }
16 }
17 class Salary extends Employee{
18     double basicSalary;
19     void getBasicSalary(double s){
20         basicSalary=s;
21     }
22 }
23 class GrossSalary extends Salary{
24     void displayDetails(){
25         double hra=basicSalary*0.2;
26         double da=basicSalary*0.1;
27         double gross=basicSalary+hra+da;
28         System.out.println("Employee ID: "+empId);
29         System.out.println("Employee Name: "+empName);
30         System.out.println("Basic Salary: "+basicSalary);
31         System.out.println("HRA: "+hra);
32         System.out.println("DA: "+da);
33         System.out.println("Gross Salary: "+gross);
34     }
35 }
```

Input

```
012  
sam  
80000
```

Output

```
Employee ID: 12  
Employee Name: sam  
Basic Salary: 80000.0  
HRA: 16000.0  
DA: 8000.0  
Gross Salary: 104000.0
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3

**10.00 / 10 marks** **Submitted**